System architecture

Logical component view based on the supplied source archive. Teal: application/data flow. Blue: authentication. Dashed: external AI service.

Synthetic cohort generator

Deterministic adult CKD records
Historical EHR + simulated IoT
Seed, version, cohort ID and UTC time

Privileged seeding scripts

Cloud Firestore

Versioned cohorts
Patient demographics and manifests

Separate evidence streams
ehr_events / iot_observations

Stored review outputs
clinical_rules / aki_predictions
Doctor review notes

Admin SDK read / write

TRUSTED SERVER BOUNDARY

FastAPI gateway

Firebase ID-token verification + exact role: doctor claim
Pydantic request validation and protected patient routes

APPLICATION SERVICES

Current clinical rules

Creatinine-only rule assessment
Observed state and evidence

AKI prediction + SHAP

Verified XGBoost pipeline
24-hour risk + model influence

Patient reports

ReportLab PDF generation
History, rules, risk and notes

Twini assistant

Project evidence retrieval
Optional synthetic patient context

Repository access stays server-side. Admin credentials and the Groq key never enter browser code.

Bearer ID token API request

JSON / PDF responses

Clinician web app

Next.js + React + TypeScript
Tailwind CSS + Recharts

Overview and monitored patients
Patient digital twin and trends
Review notes and PDF downloads
Twini conversation workspace

Sign-in / ID token

Firebase Authentication

Email/password doctor sign-in
Issues Firebase ID tokens

Context / answer

Groq API

Backend-only LLM request
Grounded answer with source labels

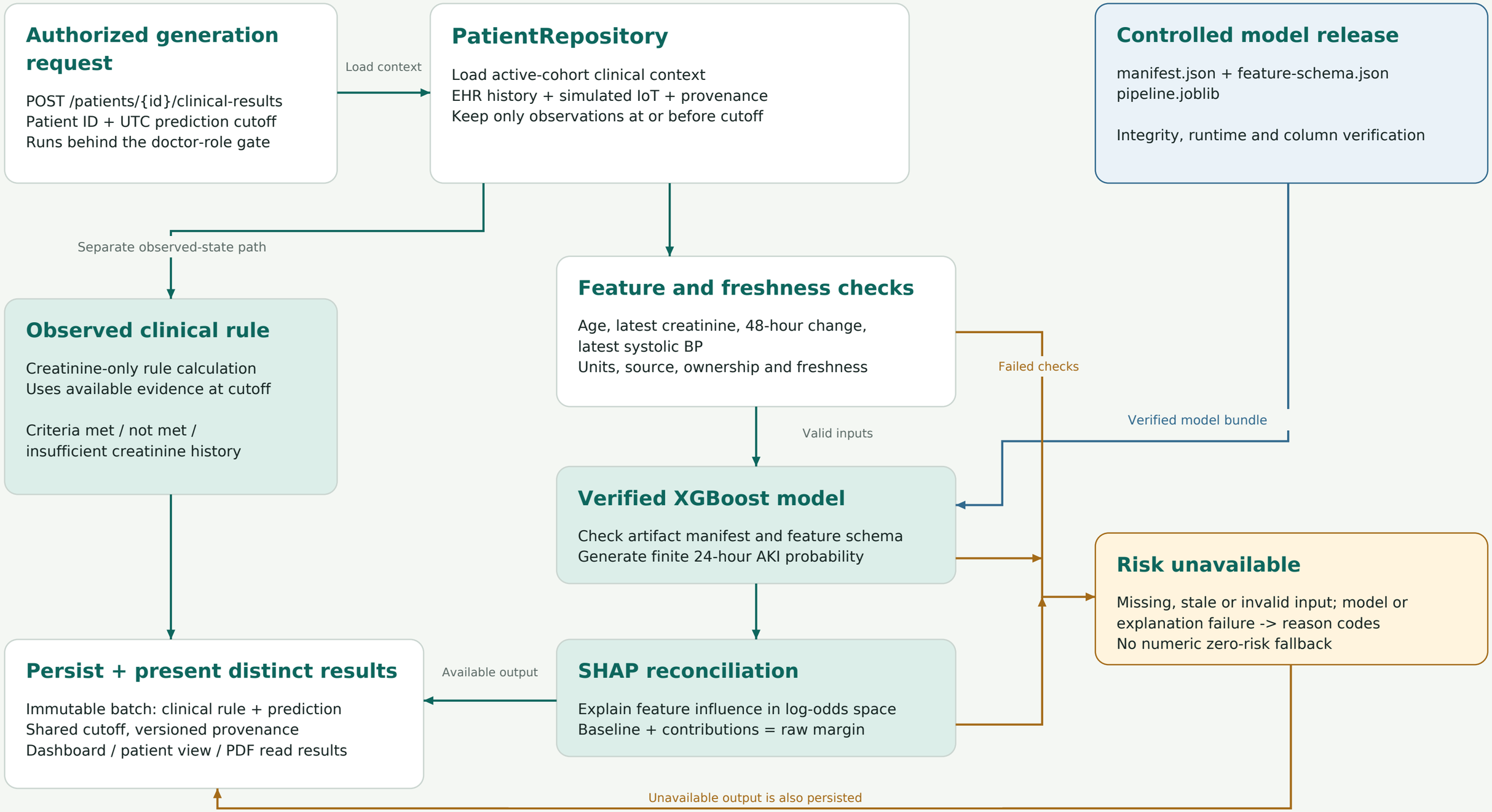
Security and data boundaries

No direct browser-to-Firestore path. EHR and IoT remain separate. Twini cannot change records or model outputs.

Source: README.md; backend/app/auth.py, patient_routes.py, patients.py, prediction_service.py, twini.py.

Prediction and evidence flow

The current clinical rule and the future prediction follow separate paths and are persisted as distinct, time-stamped results.



Read path: GET dashboard and patient endpoints retrieve stored results; viewing the dashboard does not trigger inference.